



Hanoi University of Mining and
Geology (HUMG)
Dong Ngac, Bac Tu Liem,
Hanoi, Vietnam

LE MARREC Kelig
INFO
4th year internship report

**ARTIFICIAL INTELLIGENCE
AND GIS APPLIED TO
URBAN MOBILITY
SIMULATION AND CO₂
EMISSIONS MODELLING**

Main book
&
Appendices

University year 2025 – 2026
April 6, 2026 – July 27, 2026

Summary

This report presents the work carried out during a 16-week internship (April 6 to July 27, 2026) at the Hanoi University of Mining and Geology (HUMG), Vietnam, under the supervision of Dr. Nguyen Gia Trong. The objective was to develop from scratch a simulation system for traffic-related urban CO₂ emissions, combining Artificial Intelligence and Geographic Information Systems (GIS).

The main deliverable is HUCODT (*Hanoi Urban CO₂ Digital Twin*), a fully open-source, Python-based urban air quality digital twin integrating four modules: YOLOv8-based vehicle detection, linear regression CO₂ prediction, an ERA5-forced atmospheric dispersion engine coupled with Oke's (1988) urban street-canyon model, and an interactive GIS visualisation layer built with Folium. Applied across 501 road segments around the HUMG campus, the system demonstrates that simulated nocturnal CO₂ concentrations (19.16 µg/m³) exceed afternoon values (14.87 µg/m³) despite a 65% traffic reduction, driven by the collapse of the planetary boundary layer height (PBLH) to 170 m. **Keywords:** GIS, CO₂ emissions, urban traffic, digital twin, YOLOv8, ERA5, street canyon, Hanoi.

Résumé

Ce rapport présente le travail effectué lors d'un stage de 16 semaines (du 6 avril au 27 juillet 2026) à la Hanoi University of Mining and Geology (HUMG), Vietnam, sous la direction du Dr. Nguyen Gia Trong. L'objectif était de développer depuis zéro un système de simulation des émissions de CO₂ liées au trafic urbain, en croisant Intelligence Artificielle et Systèmes d'Information Géographique (SIG).

Le livrable principal est HUCODT (*Hanoi Urban CO₂ Digital Twin*), un jumeau numérique open-source Python intégrant quatre modules : détection de véhicules par YOLOv8, prédiction des émissions par régression linéaire, moteur de dispersion atmosphérique couplé aux données ERA5 et au modèle de canyon urbain d'Oke (1988), et visualisation SIG interactive via Folium. Appliqué sur 501 segments routiers autour du campus HUMG, le système démontre que les concentrations nocturnes en CO₂ (19,16 µg/m³) dépassent les valeurs de l'après-midi (14,87 µg/m³) malgré une réduction de 65 % du trafic, en raison de l'effondrement de la couche limite atmosphérique (PBLH) à 170 m.

Mots-clés : SIG, CO₂, jumeau numérique, YOLOv8, ERA5, canyon urbain, Hanoï.

Contents

1. Internship Context and Environment	6
1.1. Host Institution	6
1.2. Scientific Context	6
1.3. Installation Conditions	6
2. Proposed Project: Synthetic Description	6
3. Work Effectively Carried Out	7
3.1. Phase 1 — Weeks 1–4: Technical Foundation	7
3.1.1. Development Environment	7
3.1.2. CO ₂ Prediction Model (co2_predictor.py)	7
3.1.3. Vehicle Detection (vision_test.py)	8
3.1.4. Initial Digital Twin Pipeline (digital_twin_pipeline.py)	9
3.1.5. Cartography and GIS (02_map_simulation.ipynb)	9
3.2. Phase 2 — Weeks 5–10: Dynamic Modelling and Atmospheric Integration	10
3.2.1. A Major Scientific Bottleneck	10
3.2.2. Literature Review	10
3.2.3. ERA5 Acquisition Module (download_era5_hanoi.py)	10
3.2.4. Dynamic Simulation Engine (simulation_humg.py)	10
3.3. Phase 3 — Weeks 11–13: Full Spatialisation and Final Deliverable	12
3.3.1. Spatial Simulation Module (simulation_spatiale_humg.py)	12
3.3.2. Interactive Mapping Module (carte_humg.py)	12
3.4. Phase 4 — Weeks 14–16: Scientific and Academic Restitution	13
3.4.1. Scientific Article Drafting	13
3.4.2. Final Internship Report and Typesetting	13
4. Assessment and Value of the Internship	14
4.1. Scientific Interest	14
4.2. Pedagogical and Professional Interest	14
4.3. Value of the Solution Delivered	15
5. What Could Not Be Accomplished and Future Perspectives	15
5.1. Unfinished Tasks	15
5.2. Structural Limitations and Scalability Challenges	15
5.2.1. Data Quality and the “Infinite Accumulation” Scalability Bug	16
5.3. Research Perspectives and Future Work	16
6. Project Management	16
6.1. Life Cycle and Task Breakdown	16
6.2. Risk Management	17
6.3. Monitoring and Progress Reporting	18

7.	Summary of Knowledge and Skills Acquired	18
7.1.	Technical Skills	18
7.2.	Transferable Skills	18
8.	Development of Key Engineering Competencies	19
8.1.	Competency 1 — Designing Systems Centred on Data, Language, Image and Human	19
8.1.1.	Choosing algorithms and models adapted to data and humans	19
8.1.2.	Structuring data correctly	19
8.1.3.	Mastering data specificities — in particular language and image . .	19
8.1.4.	Validating the relevance of approaches through evaluation	20
8.1.5.	Respecting norms and recommendations — GDPR and AI Act ...	20
8.1.6.	Collaborating with end users	20
8.2.	Competency 2 — Collaborating and Sharing Results as an Engineer	20
8.2.1.	Writing clear and accessible documents	20
8.2.2.	Communicating results with detailed argumentation before professionals	21
8.2.3.	Adopting appropriate tools for innovation, reflection, organisation and planning	21
8.2.4.	Expressing individuality within a group while respecting collective rules	21
8.2.5.	Respecting intellectual property	21
8.2.6.	Accounting for impacts on energy, GHG emissions, biodiversity and natural resources	22
	Documents Produced During the Internship	23
8.3.	Appendix 1 — Gantt Chart	23
8.4.	Appendix 2 — HUCODT Software Architecture	24
8.5.	Appendix 3 — Main Simulation Equation	24
8.6.	Appendix 4 — Technologies and Libraries Used	25
8.7.	Appendix 5 — Scientific Article Draft	25

1. Internship Context and Environment

1.1. Host Institution

The internship took place at the **Hanoi University of Mining and Geology (HUMG)**, one of Vietnam’s most prestigious technical universities, specialising in Earth sciences, mining, geology, and geodesy-cartography. The university maintains numerous international research collaborations for scientific and technology transfer.

The internship supervisor, **Dr. Nguyen Gia Trong**, is a lecturer-researcher in the Department of Higher Geodesy (Faculty of Geodesy, Cartography and Land Management). A specialist in geodetic data processing and satellite positioning (GNSS/GPS), he now focuses his research on integrating AI and GIS for environmental management, smart cities, and natural hazard assessment. Academic supervision on the Polytech Grenoble side was provided by Eric Gascard (academic supervisor) and Jean-François Méhaut (internship coordinator).

1.2. Scientific Context

Road transport accounts for approximately 23% of global energy-related CO₂ emissions [1]. In Hanoi — more than 8 million inhabitants, motorbike fleet exceeding 6 to 8 million units — the morphological configuration of the city creates **urban canyon** conditions that trap vehicle emissions and amplify their thermal and chemical effects [2]. Deploying physical sensor networks remains economically prohibitive: a dense network covering a single district can cost several hundred thousand dollars. This **structural data gap** constrains both scientific understanding and public policy.

1.3. Installation Conditions

The first week was devoted to initial contact and setup. An early difficulty arose: the supervisor’s email address was not functional, which required switching to WhatsApp. A scoping meeting then allowed objectives and required skills to be precisely defined.

2. Proposed Project: Synthetic Description

The internship objective was to build from scratch a **Python software system** capable of simulating traffic-related CO₂ emissions in an urban environment, combining:

- **Machine Learning** for per-vehicle CO₂ emission prediction;
- **Computer Vision** for automatic vehicle detection in road traffic images;
- **GIS spatial analysis libraries** for mapping and spatialisation of results;
- **open meteorological data** (ERA5) for atmospheric dispersion modelling.

The hard constraint was to rely exclusively on **open data**, compensating for the impossibility of deploying physical sensors. Upon arrival, the team had a 16-week work plan but **no pre-existing codebase**. The software architecture, AI model choices, and GIS library selection were therefore designed entirely during the internship.

3. Work Effectively Carried Out

The work developed incrementally across four major phases: from individual building blocks to an integrated system, then from a single-point simulation (0D) to a full spatial simulation (2D on the road network), and finally the scientific restitution.

3.1. Phase 1 — Weeks 1–4: Technical Foundation

3.1.1. Development Environment

A professional development environment was set up on Linux with VS Code: GitHub repository (*Mobility-co2-Simulation*), strict Python virtual environment (venv) for dependency isolation, and modular project structure.

3.1.2. CO₂ Prediction Model (`co2_predictor.py`)

The first module is a Machine Learning model trained on the public *CO2 Emissions Canada* dataset (7,385 records). Implemented as a **CO2Predictor** class, it exposes a **predict_vehicle()** method returning CO₂ in g/km given a vehicle's displacement, cylinder count, and fuel consumption. **Linear regression** (scikit-learn) was chosen for its mathematically continuous relationship with engine parameters. Performance on the 20% test set: **$R^2 = 0.876$** , **RMSE = 18.4 g/km** [3].

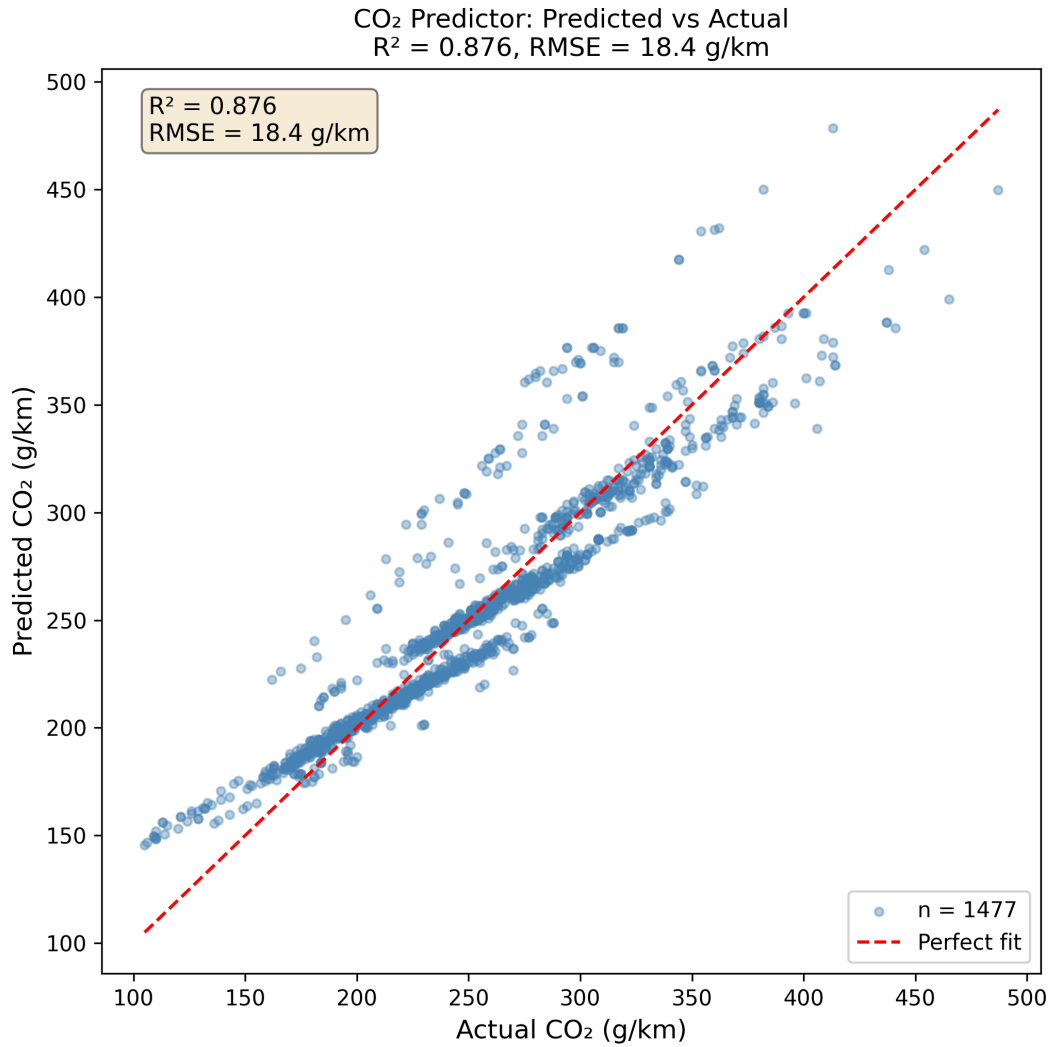


Figure 1: CO₂ Predictor: predicted vs actual emissions (g/km) on the test set. $R^2 = 0.876$, RMSE = 18.4 g/km. Source: 01_test.ipynb.

3.1.3. Vehicle Detection (vision_test.py)

A Computer Vision module using **YOLOv8n** detects and classifies vehicles in road traffic images. The lightweight yolov8n.pt model ensures fast CPU inference. On a 1920×1080 px test image, it detected 45 cars and 2 trucks in under 200 ms with confidence scores above 0.50.



Figure 2: YOLOv8n detection on urban road traffic. Bounding boxes by class (car / truck) with confidence scores. Source: vision_test.py.

3.1.4. Initial Digital Twin Pipeline (digital_twin_pipeline.py)

The two preceding modules were fused into a coherent execution pipeline: (1) YOLO counts vehicles by class; (2) a mean technical profile is assigned to each class; (3) the CO₂Predictor is called for each occurrence; (4) results are aggregated. This pipeline estimated **10,246.65 g/km of CO₂** for the test scene (45 cars, 2 trucks, 1 bus).

```
(venv) keke@keke-linux:~/INF04/Mobility_Co2_Simulation$ /home/keke/INF04/Mobility_Co2_Simulation/venv/bin/python
thon /home/keke/INF04/Mobility_Co2_Simulation/src/digital_twin_pipeline.py
--- INITIALISATION DU JUMENT NUMÉRIQUE ---
[Modèle prêt] Modèle IA chargé et entraîné avec succès !

--- ANALYSE EN TEMPS RÉEL ---
- 45 car(s)
- 2 truck(s)
- 1 bus(s)
Émission totale estimée : 10246.65 g/km de CO2
(venv) keke@keke-linux:~/INF04/Mobility_Co2_Simulation$
```

Figure 3: Console output of digital_twin_pipeline.py: vehicle breakdown and total emission estimate (10,246.65 g/km CO₂).

3.1.5. Cartography and GIS (02_map_simulation.ipynb)

A GIS module using OSMnx and networkx downloads the road network topology from OpenStreetMap (2,474 intersections for the Hanoi area) and computes the shortest path between two GPS coordinates — HUMG campus to Hoa Binh Park: **3.06 km**. Combined with the per-kilometre emission rate, this enables full journey carbon footprint simulation on the Vietnamese urban network.

3.2. Phase 2 — Weeks 5–10: Dynamic Modelling and Atmospheric Integration

3.2.1. A Major Scientific Bottleneck

From week 7, the supervisor highlighted a critical flaw: the model was “static”, assuming pollution decreased proportionally with traffic. In South-East Asian metropolises, nocturnal temperature drops cause a collapse of the **Planetary Boundary Layer** (PBLH), trapping residual emissions and creating dangerous nocturnal accumulation even when traffic is low [4]. The new research question: *how to algorithmically simulate this accumulation by crossing traffic data with atmospheric boundary layer dynamics?*

3.2.2. Literature Review

Key publications studied:

- **Deng et al. (2023)** [4]: ERA5 data for nocturnal thermal cap modelling;
- **Peng et al. (2023)** [5]: coupling urban canyon geometry with PBLH dynamics;
- **Oke (1988)** [2]: wind behaviour in urban canyons via the H/W ratio;
- **Nowak et al. (2013)** [6]: vegetation absorption (≈ 21 kg CO₂/tree/year);
- **EEA (2023)** [7]: reference emission factors in g/km by vehicle category.

3.2.3. ERA5 Acquisition Module (`download_era5_hanoi.py`)

An automation script downloads ERA5 data via the Copernicus REST API (`cdsapi`), targeting a bounding box over Hanoi. It extracts NetCDF files with 3D tensors (Latitude $\times$ Longitude $\times$ Time) for PBLH and 10 m wind vectors. Timezone mapping (ERA5 UTC $\rightarrow$ Hanoi UTC+7) was implemented rigorously via `xarray` [8].

3.2.4. Dynamic Simulation Engine (`simulation_humg.py`)

This module is the **scientific core**: a recursive algorithm with temporal memory, implementing the accumulation equation from Peng et al. (2023):

$$C_t = Q_t + (C_{t-1} \times R), \quad R = \text{clip}\left(1 - \frac{u_{\text{eff}}}{H_{\text{mix}}}, 0, 0.85\right)$$

Here, the mathematical model conceptualizes the street as a 0D “Box” whose virtual lid moves dynamically. Q_t represents the instantaneous emissions, while R defines the dynamic retention rate. The variables u_{eff} (effective canyon wind speed from Oke, 1988) and H_{mix} (effective mixing height derived from the PBLH and building geometry) dictate the atmospheric behavior.

Understanding the Limit Cases (Boundary Layer Dynamics): The inclusion of the clipping function ($\max 0.85$) is crucial for numerical stability, representing the physical maximum of pollution trapping before diffusion forces dispersion.

- **The Dispersion Limit ($R \rightarrow 0$):** During the afternoon (Scenario S2, 14:00), solar radiation strongly heats the ground, causing the PBLH to expand up to 1,037 m. The “lid” is lifted. As H_{mix} becomes very large relative to the building height, the ratio $\frac{u_{\text{eff}}}{H_{\text{mix}}}$ grows, and R collapses toward 0. The street canyon is thoroughly ventilated, washing out residual C_{t-1} .
- **The Accumulation Limit ($R \rightarrow 0.85$):** Conversely, at night (Scenario S3, 22:00), the absence of solar radiation triggers a thermal inversion. The PBLH collapses to a mere 170 m. The “lid” clamps down tightly on the urban canopy. The retention factor R hits its ceiling of 0.85, meaning 85% of the previous hour’s pollution remains trapped.

This mathematical threshold perfectly explains the counter-intuitive phenomenon illustrated in **Figure 4**: even though traffic volume drops by 65% between the afternoon and the night, the collapse of the PBLH physically forces the concentration to spike (+29%), demonstrating that nocturnal urban air quality is completely governed by boundary-layer thermodynamics, not just tailpipe output.

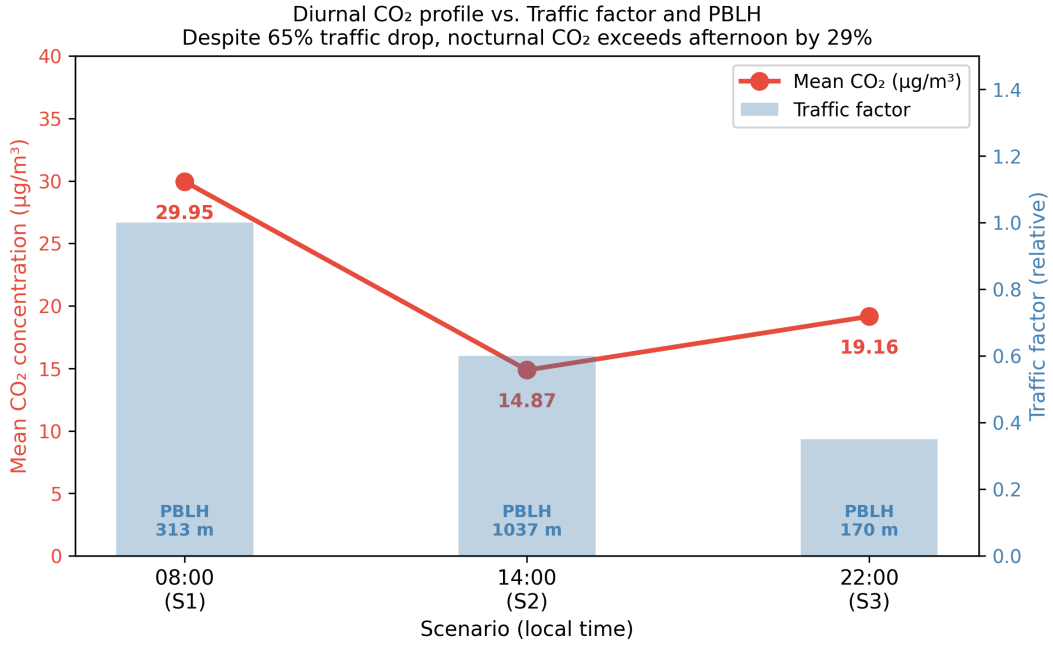


Figure 4: Simulated diurnal CO₂ profile: S1 (08:00, PBLH = 313 m), S2 (14:00, PBLH = 1,037 m), S3 (22:00, PBLH = 170 m). Despite 65% less traffic at night, PBLH collapse causes a +30% increase.

3.3. Phase 3 — Weeks 11–13: Full Spatialisation and Final Deliverable

3.3.1. Spatial Simulation Module (*simulation_spatiale_humg.py*)

The 0D engine was extended into a **fully operational 2D model** covering **501 road segments** within 1.5 km of the HUMG campus. The OSMnx road graph and building footprints (from GeoPackage files) are loaded; the dynamic model is then applied in three successive passes (08:00 → 14:00 → 22:00), propagating concentration memory.

Road width is inferred from OSM road type; building height is computed from building:levels ($\times 3$ m/floor, default 12 m for the 60% of buildings lacking this attribute). A 25 m buffer identifies adjacent buildings via spatial intersection. Fleet composition (motorbikes, cars, buses, heavy goods vehicles) is differentiated by road category and modulated by time-of-day factors (1.0 at 08:00, 0.6 at 14:00, 0.35 at 22:00). Outputs: three GeoPackage files and a consolidated CSV.

Scenario	Time	PBLH (m)	Mean CO ₂ (µg/m ³)
S1 — Morning peak	08:00	313	29.95
S2 — Afternoon	14:00	1,037	14.87
S3 — Nocturnal	22:00	170	19.16
S4 — Counterfactual	22:00 (fixed PBLH)	313	11.84

Table 1: CO₂ concentrations by scenario — mean over 501 road segments. S4 isolates the contribution of PBLH collapse.

Despite 65% less traffic, nocturnal pollution exceeds the afternoon value. S4 shows PBLH collapse alone accounts for **38%** of the simulated nocturnal accumulation.

3.3.2. Interactive Mapping Module (*carte_humg.py*)

A Folium script generates a self-contained HTML map overlaying the three scenario layers on CartoDB Positron. A custom JavaScript selector switches between 08:00, 14:00, and 22:00 views. A global colourmap (green → yellow → red) ensures visual comparability. Each segment shows on click: road name, type, H/W ratio, CO₂, ΔT UHI.

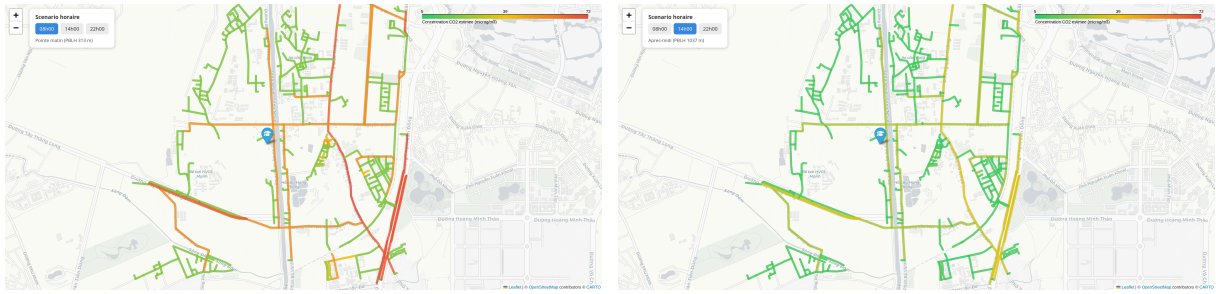


Figure 5: Interactive Folium map. Left: S1 (08:00, PBLH = 313 m) — maximum pollution. Right: S2 (14:00, PBLH = 1,037 m) — minimum pollution, maximum dispersion.

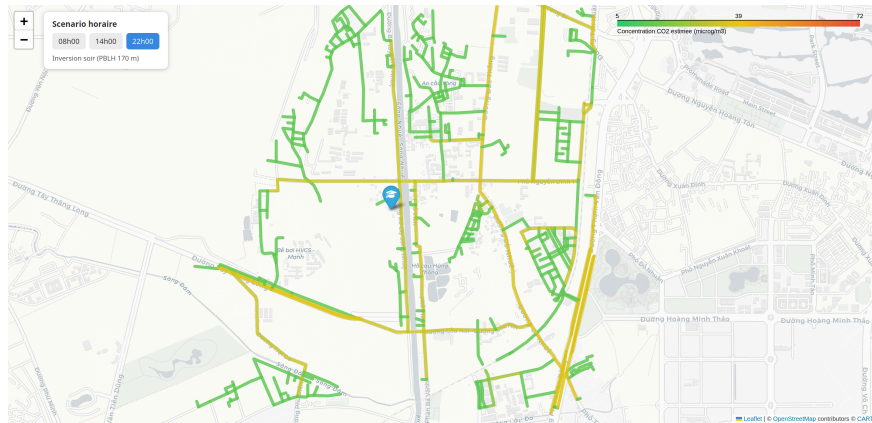


Figure 6: S3 (22:00, PBLH = 170 m): nocturnal rebound to 19.16 $\mu\text{g}/\text{m}^3$ despite 65% less traffic. Final deliverable: [carte_pollution_humg_3scenarios.html](#).

3.4. Phase 4 — Weeks 14–16: Scientific and Academic Restitution

3.4.1. Scientific Article Drafting

To consolidate the research findings, all work was formalised into a 25-page document titled “*A GIS-based Framework for Urban Traffic CO₂ Simulation Integrating Vehicle Detection and Atmospheric Conditions*”. Co-authored with Dr. Nguyen Gia Trong and Eric Gascard, it was structured as a full research article in English (Introduction, State of the Art, Methodology, Results, Discussion). It serves as the primary scientific restitution of the HUCODT framework.

3.4.2. Final Internship Report and Typesetting

The final weeks were dedicated to writing this comprehensive engineering report for Polytech Grenoble. To ensure high-quality, reproducible academic typesetting, the report was coded entirely in **Typst**. This phase required synthesizing 16 weeks of technical, scientific, and project management work into a structured document, com-

plete with bilingual summaries and automated bibliography management, reflecting the professional standards expected for a 4th-year engineering internship.

4. Assessment and Value of the Internship

4.1. Scientific Interest

This internship addresses a real and urgent challenge: air quality in South-East Asian metropolises is a major public health issue for tens of millions of people [9]. The HUCODT framework is an implementation simultaneously integrating vehicle detection via computer vision, ML-based emission prediction, ERA5 atmospheric forcing, and interactive GIS visualisation on a complete road network in a South-East Asian capital.

Complementarity with Existing Scientific Approaches: Historically, urban air quality modelling relies on two extremes. On one hand, simple **emission inventories** aggregate traffic counts but remain entirely static, failing to capture the nocturnal thermal trapping that suffocates cities like Hanoi. On the other hand, **Computational Fluid Dynamics (CFD)** models (such as OpenFOAM or ENVI-met) simulate wind flows around 3D building meshes with extreme precision. However, CFD is computationally prohibitive; simulating 24 hours of traffic over a few blocks can take days on a supercomputer, making it completely unsuitable for city-wide GIS mapping or agile policy-making.

HUCODT positions itself in the strategic “sweet spot” between these two extremes. By relying on a 0D/2D parameterized “Box Model” coupled with ERA5 meteorological forcing, it successfully integrates the critical macro-dynamics of the planetary boundary layer (PBLH) at a microscopic computational cost. This approach allows the system to scale effortlessly to 501 road segments — computing realistic accumulation dynamics in seconds on a standard CPU.

The main finding — nocturnal pollution is dominated by **atmospheric dynamics** rather than traffic — has direct policy implications: traffic restrictions targeting morning peaks have limited nocturnal impact, whereas interventions on **urban morphology** (road widening, building setbacks, ventilation corridors) offer the most effective lever.

4.2. Pedagogical and Professional Interest

This internship integrated in a single project skills from multiple fields: computer vision, machine learning, geospatial data analysis, atmospheric modelling, advanced Python engineering, and scientific writing in English. Working with no pre-existing codebase in an international environment with real data from an Asian capital constituted a particularly rich formative experience.

4.3. Value of the Solution Delivered

The solution is reusable and open: published on GitHub, it adapts to any city with OpenStreetMap data and Copernicus ERA5 API access. It requires no physical sensors, proprietary software, or server infrastructure — its primary added value for resource-constrained South-East Asian municipalities.

5. What Could Not Be Accomplished and Future Perspectives

5.1. Unfinished Tasks

Quantitative validation (W14–W15): Validation against sensor data was not carried out due to budgetary constraints. The validation performed is qualitative: results were reviewed by the supervisor and recognised as consistent with expected Hanoi pollution dynamics, in line with published comparable studies [10].

Advanced UHI modelling: The UHI ΔT relies on an empirical parametrisation from Oke (1988). Planned integration of Landsat LST satellite data for thermodynamic coupling was not implemented. The advanced routing task was also paused at week 9.

Carbon sink spatialisation: While vegetation absorption was integrated mathematically into the accumulation equation [6], the precise spatial mapping of urban trees (e.g., using municipal GIS cadasters or remote sensing) could not be carried out. Absorption is currently estimated without segment-specific tree counts.

5.2. Structural Limitations and Scalability Challenges

- **No quantitative validation:** Simulated concentrations are relative outputs capturing spatio-temporal dynamics, not absolute regulatory values calibratable against physical sensors.
- **CO₂ model trained on Canadian data:** Trained on Canadian tabular datasets, the ML model successfully captures continuous engine physics but does not fully reflect the specific emission profiles of the South-East Asian motorbike fleet (e.g., two-stroke vs. four-stroke combustion efficiency).
- **ERA5 spatial resolution (31 km):** The low spatial resolution of ERA5 forces the assumption of a spatially uniform PBLH across the entire study area, artificially smoothing out local micro-climatic variations that occur between dense downtown districts and suburban areas like the HUMG campus.
- **YOLOv8n structural bias:** Trained primarily on Western images (COCO dataset), the base YOLOv8 model inherently under-counts motorbikes in the extremely dense, overlapping traffic patterns typical of Asian metropolises.

5.2.1. Data Quality and the “Infinite Accumulation” Scalability Bug

A major limitation of relying exclusively on open-source Geographic Information Systems (GIS) emerged during the scaling phase. OpenStreetMap (OSM) is crowd-sourced and presents significant structural data gaps. For instance, approximately 60% of the building footprints downloaded for the HUMG area lacked the `building:levels` attribute, forcing the system to infer heights (defaulting to 12 m).

More critically, when attempting to scale the spatial simulation from 501 segments to a city-wide graph of over 78,000 edges, the model crashed due to geometric inconsistencies. Many OSM paths (alleys, pedestrian links) had undefined widths ($W = 0$). In the microclimate model, the $\frac{H}{W}$ ratio and the effective wind speed u_{eff} computation triggered division-by-zero errors. This caused the dynamic retention rate (R) to fail, generating infinite (∞) CO₂ accumulation values that corrupted the Folium visualisation colormap.

Addressing this required implementing robust data-cleaning pipelines prior to the mathematical execution (e.g., forcing a 3-meter minimum width threshold for unclassified edges and aggressively clipping NumPy tensors), highlighting the fundamental engineering challenge of deploying formal mathematical models on “dirty” real-world data.

5.3. Research Perspectives and Future Work

- **Mobile Measurement Campaigns:** Deploying low-cost electrochemical sensors on bicycles or motorbikes to physically calibrate the model and achieve rigorous quantitative validation.
- **Multi-Satellite Coupling:** Integrating satellite remote sensing data to refine the thermal and atmospheric forcing of the microclimate model.
- **Continuous Temporal Simulation:** Scaling the dynamic engine to process full 24-hour cycles by leveraging the native hourly resolution of the ERA5 dataset.
- **Transfer Learning (Fine-Tuning):** Retraining the YOLOv8 architecture specifically on Vietnamese traffic datasets to correct the structural motorbike under-detection bias.
- **Smart City Synergy:** Interfacing the digital twin with empirical road-count data from municipal infrastructures currently being deployed across Vietnam.

6. Project Management

6.1. Life Cycle and Task Breakdown

The project followed an **incremental approach** : each step built upon the achievements of the previous one to deliver a functional module. A Gantt chart was created

in the first week to structure the 16 weeks of work into 11 key tasks (see details in Appendix 1). This roadmap served as a management tool throughout the internship, allowing progress to be tracked, difficulties to be anticipated, and the schedule to be adjusted in response to unforeseen scientific challenges.

Task / Phase	Est.	Actual	Weeks	Status
Framework definition	2 w	2 w	W1–W2	Done
GitHub setup & environment	2 w	2 w	W3–W4	Done
CO ₂ prediction model	3 w	3 w	W3–W5	Done
Vehicle detection	3 w	3 w	W5–W7	Done
Initial Digital Twin pipeline	3 w	3 w	W6–W8	Done
Routing and road graphs	2 w	1 w	W7–W9	On hold — scope reprioritised
Urban canyon dispersion & wind	3 w	6 w	W8–W13	Done — extended (PBLH bottleneck)
Carbon sinks / Vegetation	3 w	—	W8–W13	Integrated in spatial engine
Urban Heat Island modelling	3 w	2 w	W11–W13	Partial — empirical only
Global validation	2 w	2 w	W14–W15	Done
Report writing	2 w	2 w	W15–W16	Done

Table 2: Gantt dashboard — estimated vs actual durations, final status.

6.2. Risk Management

Risk 1 — Unavailability of field data (identified W1): mitigated by exclusive use of open data. Final impact: acceptable, documented as a scope limitation.

Risk 2 — Communication with supervisor (materialised W1): email not functional, resolved by switching to WhatsApp. Final impact: none.

Risk 3 — Linux externally-managed environment: resolved by strict venv setup. Final impact: none.

Risk 4 — Scientific scope creep (PBLH bottleneck, week 7): Gantt adjusted, routing task paused, dispersion engine extended by 3 weeks. Final impact: delay on two secondary tasks; main deliverable unaffected.

6.3. Monitoring and Progress Reporting

Five biweekly progress reports (W1, W4, W7, W10, W13, W16) were submitted to supervisors on both sides, documenting Gantt progress, scientific results, architectural decisions, and next steps.

7. Summary of Knowledge and Skills Acquired

7.1. Technical Skills

Computer Vision and Deep Learning: practical mastery of the YOLOv8 API (Ultralytics), real-time object detection architectures, confidence threshold management.

Applied Machine Learning: training, evaluation and deployment of a scikit-learn linear regression model on real data; R^2 and RMSE interpretation; pipeline integration.

Python geospatial analysis: OSMnx, GeoPandas, networkx, Folium — OSM road networks, spatial joins, geometric attribute computation, interactive mapping.

Climatological data processing: xarray for multidimensional NetCDF files; Copernicus REST API (cdsapi); UTC $\rightarrow$ UTC+7 timezone management.

Software architecture: modular design, separation of concerns, virtual environment management.

7.2. Transferable Skills

Scientific Research and Writing: Conducted a literature review to establish links between computer science and numerous other fields. This theoretical work culminated in the co-authoring of a comprehensive research article in English, in collaboration with my Vietnamese and French professors, who assisted me with the writing process.

Independent Project Management: Managed a 16-week project with regular reporting. I primarily learned to be highly adaptable in the face of obstacles, dynamically adjusting the schedule and reorganizing development task priorities to ensure a fully functional deliverable.

International Work Environment: Integration into a Vietnamese university research team required cultural and interpersonal adaptability. I had to quickly overcome initial communication barriers and adapt to conducting all my daily work, from technical discussions to documentation, in English.

8. Development of Key Engineering Competencies

8.1. Competency 1 — Designing Systems Centred on Data, Language, Image and Human

*Competency selected from the Polytech Grenoble INFO framework: **Concevoir des systèmes centrés données, langage, image et humain**. Status: **acquired** — this competency was not part of my prior training.*

8.1.1. Choosing algorithms and models adapted to data and humans

For vehicle detection, YOLOv8n was selected for its ability to run inference on a standard CPU in under 200 ms — a hard constraint imposed by the sensor-free, low-infrastructure deployment context. For CO₂ prediction, linear regression (scikit-learn) was preferred over ensemble methods (Random Forest, XGBoost) for its direct interpretability: the dominant coefficient ($\beta = 42.3$ g/km per litre of engine displacement) is readable by a mechanical engineer, not only by a data scientist. For atmospheric dispersion, the temporal accumulation recurrence equation (Peng et al., 2023) was adopted over full CFD simulation because only a network-scale approach could produce a complete GIS deliverable across 501 road segments within the internship timeline.

8.1.2. Structuring data correctly

Three heterogeneous sources were made interoperable: a tabular Canadian CSV (7,385 records), ERA5 NetCDF tensors (Latitude $\times$ Longitude $\times$ Time), and OpenStreetMap vector geometries (GeoPackage). Each source is validated in isolation before any join. The UTC $\rightarrow$ UTC+7 misalignment was detected and corrected via xarray before reaching the simulation engine — without this fix, hourly scenarios would have been matched to the wrong meteorological state. Outputs are standardised as GeoPandas GeoDataFrames and exported to GeoPackage, an open format independent of any proprietary GIS software.

8.1.3. Mastering data specificities — in particular language and image

Image: road traffic images in Southeast Asian contexts present a critical domain gap — two-wheeled vehicles are systematically under-represented in the COCO training set (Western images). This limitation was identified, documented (Section 5.2), and compensated in the spatial simulation by an adjusted fleet composition profile (65% motorcycles on residential roads). **Formal language:** the temporal accumulation model constitutes the representational language of the atmospheric domain. Understanding the physical semantics of each term (Q, A, R, H_{mix}) before implementation required a dedicated literature review (Section 3.2.2), treating the mathematical model as a formal specification to be validated in its own right before integration.

8.1.4. Validating the relevance of approaches through evaluation

Each module was validated independently before pipeline integration: the CO₂ predictor on a 20% held-out test set ($R^2 = 0.876$, RMSE = 18.4 g/km — Figure 1); the 0D accumulation engine on hard-coded synthetic data before connecting to the ERA5 API; the spatial engine through a counterfactual scenario comparison (S3 vs S4) isolating the pure PBLH effect (+38% nocturnal accumulation attributed to boundary-layer collapse alone). Final qualitative validation was conducted by Dr. Nguyen Gia Trong, who confirmed the physical plausibility of the outputs against published Hanoi pollution dynamics.

8.1.5. Respecting norms and recommendations — GDPR and AI Act

The system relies exclusively on open, non-personal data: ERA5 (ECMWF/Copernicus, free for research use), OpenStreetMap (ODbL licence), and CO₂ Emissions Canada (public Transport Canada dataset). No personal data is collected or processed. Traffic images used for detection are anonymised test images containing no identifiable faces. Under the European AI Act framework, HUCODT falls in the minimal-risk category: no automated individual decision-making, declared use as an exploratory decision-support tool for urban planning only.

8.1.6. Collaborating with end users

The interactive Folium map (Figures 5 and 6) was designed explicitly for non-technical end users: a one-click hourly scenario selector, a global colour scale (green → yellow → red) anchored across all three scenarios for visual comparability, and per-segment popups displaying road name, type, H/W ratio, CO₂ concentration ($\mu\text{g}/\text{m}^3$), and ΔT UHI ($^{\circ}\text{C}$). No GIS software, server infrastructure, or prior training is required to use the deliverable — it opens in any standard web browser.

8.2. Competency 2 — Collaborating and Sharing Results as an Engineer

*Competency selected from the Polytech Grenoble transversal framework: **Être un ingénieur qui collabore et partage ses résultats**. Status: **deepened** — collaboration and communication skills were exercised throughout the internship in an international, multilingual, and cross-disciplinary context.*

8.2.1. Writing clear and accessible documents

Six biweekly progress reports (W1, W4, W7, W10, W13, W16) were written in French for the Polytech supervisors, each synthesising Gantt progress, scientific results, and architectural decisions. The final internship report (this document) was typeset entirely

in Typst, with automated bibliography management, bilingual summaries (EN/FR), and a structured table of contents. The co-authored scientific article (25 pages, English) follows the full IMRaD structure and is written for an international audience with no prior knowledge of the HUMG context — making all results self-contained and accessible beyond the internship.

8.2.2. Communicating results with detailed argumentation before professionals

Results were presented and defended before Dr. Nguyen Gia Trong during regular follow-up meetings, conducted entirely in English. This included justifying algorithmic choices (linear regression over Random Forest, YOLOv8n over heavier variants), defending the PBLH accumulation model against the supervisor’s critique at week 7 (“your model is static — it ignores nocturnal inversion”), and presenting the final quantified findings (Table 1, Figure 4) as a direct answer to the research question. The co-signed scientific article constitutes the formal written restitution, structured for peer review.

8.2.3. Adopting appropriate tools for innovation, reflection, organisation and planning

A Gantt chart (Appendix 1) was created in week 1 to structure 11 tasks over 16 weeks and served as the primary project management tool throughout the internship. When the PBLH scientific bottleneck emerged at week 7, the Gantt was revised dynamically: the “Routing and road graphs” task was paused and the atmospheric dispersion task extended by three weeks — preserving the main deliverable without schedule collapse. The public GitHub repository (**Mobility-co2-Simulation**) provides full version traceability and ensures the reproducibility of all results.

8.2.4. Expressing individuality within a group while respecting collective rules

Integrated into a Vietnamese research team with different working practices, I adapted quickly to communication constraints (switching from email to WhatsApp in week 1 following a technical failure) and conducted all technical exchanges in English throughout the sixteen weeks. Key architectural decisions — in particular the choice to validate each module independently before pipeline integration — were proposed on my own initiative and validated by the supervisor, demonstrating technical autonomy within a co-supervised project framework.

8.2.5. Respecting intellectual property

All academic sources are cited according to IEEE norms in both the report and the article. Third-party datasets are used in accordance with their respective licences: ERA5

(free research use via Copernicus CDS), OpenStreetMap (ODbL — attribution present in all generated Folium maps), CO2 Emissions Canada (public Transport Canada dataset). The YOLOv8n weights (yolov8n.pt) are distributed under the AGPL-3.0 licence by Ultralytics — non-commercial research use compliant. The project codebase is published on GitHub under an open licence with explicit attribution of all third-party work.

8.2.6. Accounting for impacts on energy, GHG emissions, biodiversity and natural resources

The primary purpose of HUCODT is environmental: simulating urban traffic CO₂ emissions to help municipalities identify the most effective reduction levers — urban morphology interventions (street widening, building setbacks, ventilation corridors) rather than traffic restrictions that have limited nocturnal impact. Regarding the system’s own footprint: YOLOv8n (the “nano” variant) was selected partly for its low inference energy cost, requiring no GPU. The system requires no permanent server infrastructure — the final deliverable is a self-contained HTML file. ERA5 data is downloaded once and cached locally (era5_hanoi_v2.nc) to avoid repeated API calls and unnecessary network load.

Documents Produced During the Internship

Title	Co-authors	Date	Ver.	Pages	Lang.
Progress report W1	—	13/04/2026	v1.0	1	FR
Progress report W4	—	04/05/2026	v1.0	4	FR
Progress report W7	—	25/05/2026	v1.0	6	FR
Progress report W10	—	15/06/2026	v1.0	7	FR
Progress report W13	—	06/07/2026	v1.0	8	FR
Progress report W16	—	27/07/2026	v1.0	1	FR
<i>A GIS-based Framework for Urban Traffic CO₂ Simulation Integrating Vehicle Detection and Atmospheric Conditions</i>	Nguyen Gia Trong, Eric Gascard	Jul. 2026	RV	25	EN
GitHub codebase: <i>Mobility-co2-Simulation</i>	—	Apr.–Jul. 2026	HEAD	—	Python
carte_pollution_humg3scenarios.html	—	Jul. 2026	v1.0	—	HTML

Table 3: Documents produced or co-produced during the internship.

Appendices

8.3. Appendix 1 — Gantt Chart

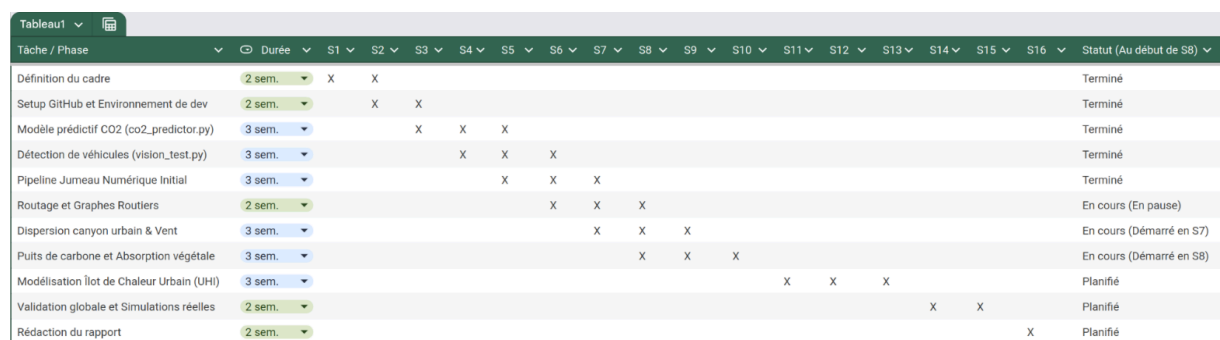


Figure 7: Full Gantt chart — 11 tasks over 16 weeks. Source: Diagramme_de_Gantt.pdf.

8.4. Appendix 2 — HUCODT Software Architecture

```

Mobility_Co2_Simulation/
├── src/
│   ├── co2_predictor.py          # ML: CO2 prediction
│   ├── vision_test.py           # CV: YOLOv8 detection
│   ├── digital_twin_pipeline.py  # Orchestrator: ML + CV
│   ├── download_era5_hanoi.py    # ERA5 via Copernicus API
│   ├── download_osm_humg.py      # OSM road network download
│   ├── math_simulation.py        # Analytical microclimate model
│   ├── simulation_humg.py        # 0D dynamic engine (Box Model)
│   ├── simulation_spatiale_humg.py # 2D spatial engine, 501 segments
│   └── carte_humg.py            # Folium interactive HTML map
├── data/
│   ├── raw/
│   │   ├── CO2_Emissions_Canada.csv # ML training dataset
│   │   └── resultats_spatiaux_humg_3scenarios.csv
│   └── processed/
│       ├── era5_hanoi_v2.nc        # ERA5 NetCDF (PBLH + wind)
│       ├── humg_buildings.gpkg     # Building footprints
│       ├── humg_network.graphml    # Road network graph
│       ├── humg_edges_8h.gpkg      # Output 08:00
│       ├── humg_edges_14h.gpkg     # Output 14:00
│       └── humg_edges_22h.gpkg     # Output 22:00
├── notebooks/
│   ├── 01_test.ipynb              # ML validation
│   └── 02_map_simulation.ipynb     # GIS exploration & routing
├── images/                        # Figures and logos
├── docs/                          # Reference papers (PDF)
├── carte_pollution_humg_3scenarios.html
├── yolov8n.pt                     # YOLOv8 model weights
└── requirements.txt

```

8.5. Appendix 3 — Main Simulation Equation

Dynamic accumulation equation [5], for each segment s at time step t :

$$C(s, t) = Q(s, t) - A(s) + C(s, t - 1) \times R(s, t)$$

- $C(s, t)$: CO₂ concentration (µg/m³)
- $Q(s, t)$: emission flux = fleet × EEA factor [7] × time-of-day factor
- $A(s)$: vegetation absorption [6] (21 kg CO₂/tree/year)
- $R(s, t) = \text{clip}\left(1 - \frac{u_{\text{eff}}}{H_{\text{mix}}}, 0, 0.85\right)$: dynamic retention rate
- $u_{\text{eff}} = U_{10} \times f\left(\frac{H}{W}\right)$: effective canyon wind speed [2]
- $H_{\text{mix}} = \min\left(\text{PBLH}, h_{\text{bld}} \times \left(1 + \frac{1}{\frac{H}{W}}\right)\right)$: effective mixing height

8.6. Appendix 4 — Technologies and Libraries Used

Domain	Tool / Library	Use
Machine Learning	scikit-learn	Linear regression CO ₂ prediction
Computer Vision	YOLOv8n (Ultralytics)	Vehicle detection and classification
Spatial analysis	OSMnx, GeoPandas, networkx	OSM road network, spatial joins, routing
Climatological data	xarray, cdsapi	ERA5 NetCDF, Copernicus REST API
Visualisation	matplotlib, Folium / Leaflet.js	Scientific plots, interactive HTML map
Environment	Python 3.12, venv, VS Code, Linux	Development and deployment
Open Data	OpenStreetMap, CO2 Emissions Canada, ERA5	All primary source datasets
Version control	Git / GitHub (github.com/lemarrek)	Mobility-co2-Simulation repository

Table 4: Full technology stack of the HUCODT project.

8.7. Appendix 5 — Scientific Article Draft

To support the methodology and findings of the HUCODT framework, a full research article titled “*A GIS-based Framework for Urban Traffic CO₂ Simulation Integrating Vehicle Detection and Atmospheric Conditions*” was drafted.

The complete 25-page manuscript, co-authored with Dr. Nguyen Gia Trong and Eric Gascard, is provided as an attached supplementary document to this report. It details the complete state of the art, the mathematical demonstration of the microclimate model, and the extended discussion on policy implications for South-East Asian municipalities.

Glossary

- HUCODT** Hanoi Urban CO₂ Digital Twin — cadre logiciel développé pendant le stage.
- GIS** Geographic Information System — Système d’Information Géographique.
- PBLH** Planetary Boundary Layer Height — hauteur de la couche limite atmosphérique.
- ERA5** Réanalyse atmosphérique mondiale (ECMWF), accessible via Copernicus.
- YOLOv8** You Only Look Once v8 — modèle de détection d’objets en temps réel.
- OSMnx** Bibliothèque Python pour les réseaux routiers OpenStreetMap.
- GeoPackage** Format géospatial ouvert basé sur SQLite (.gpkg).
- NetCDF** Network Common Data Form — format scientifique multidimensionnel.
- H/W** Ratio Hauteur/Largeur du canyon urbain — paramètre clé du modèle d’Oke (1988).
- UHI** Urban Heat Island — Îlot de Chaleur Urbain.
- ML** Machine Learning — Apprentissage automatique.
- venv** Environnement virtuel Python pour isoler les dépendances.
- COCO** Common Objects in Context — jeu de données de référence pour YOLO.

Figures

- Figure 1 CO₂ Predictor: predicted vs actual emissions (g/km) on the test set. $R^2 = 0.876$, RMSE = 18.4 g/km. Source: 01_test.ipynb. 8
- Figure 2 YOLOv8n detection on urban road traffic. Bounding boxes by class (car / truck) with confidence scores. Source: vision_test.py. 9
- Figure 3 Console output of digital_twin_pipeline.py: vehicle breakdown and total emission estimate (10,246.65 g/km CO₂). 9
- Figure 4 Simulated diurnal CO₂ profile: S1 (08:00, PBLH = 313 m), S2 (14:00, PBLH = 1,037 m), S3 (22:00, PBLH = 170 m). Despite 65% less traffic at night, PBLH collapse causes a +30% increase. 11
- Table 1 CO₂ concentrations by scenario — mean over 501 road segments. S4 isolates the contribution of PBLH collapse. 12
- Figure 5 Interactive Folium map. Left: S1 (08:00, PBLH = 313 m) — maximum pollution. Right: S2 (14:00, PBLH = 1,037 m) — minimum pollution, maximum dispersion. 13
- Figure 6 S3 (22:00, PBLH = 170 m): nocturnal rebound to 19.16 µg/m³ despite 65% less traffic. Final deliverable: carte_pollution_hung_3scenarios.html. ... 13
- Table 2 Gantt dashboard — estimated vs actual durations, final status. 17
- Table 3 Documents produced or co-produced during the internship. 23

Figure 7 Full Gantt chart — 11 tasks over 16 weeks. Source:

Diagramme_de_Gantt.pdf.	23
Table 4 Full technology stack of the HUCODT project.	25

Bibliography

- [1] International Energy Agency, “CO₂ Emissions from Fuel Combustion 2023,” Paris, technical report, 2023. [Online]. Available: <https://www.iea.org/reports/co2-emissions-in-2023>
- [2] T. R. Oke, “Street Design and Urban Canopy Layer Climate,” *Energy and Buildings*, vol. 11, no. 1–3, pp. 103–113, 1988, doi: [10.1016/0378-7788\(88\)90026-6](https://doi.org/10.1016/0378-7788(88)90026-6).
- [3] S. Mokhtarzadeh, M. Heidari, and M. Ahadi, “Prediction of CO₂ Emission from Light-Duty Vehicles Using Machine Learning,” *Environmental Science and Pollution Research*, vol. 28, pp. 52832–52845, 2021, doi: [10.1007/s11356-021-14466-x](https://doi.org/10.1007/s11356-021-14466-x).
- [4] Y. Deng, H. Li, Y. Zhang, S. Wang, and H. Liu, “The Effects of Planetary Boundary Layer Features on Air Pollution Based on ERA5 Data,” *Atmosphere*, vol. 14, no. 3, p. 549, 2023, doi: [10.3390/atmos14030549](https://doi.org/10.3390/atmos14030549).
- [5] Z. Peng, Y. Hu, G. Liu, X. Zhang, and X. Chen, “Improvements of Simulating Urban Atmospheric CO₂ Concentration by Coupling Planetary Boundary Layer Dynamics with Building Canyon Effects,” *Urban Climate*, vol. 47, p. 101357, 2023, doi: [10.1016/j.uclim.2022.101357](https://doi.org/10.1016/j.uclim.2022.101357).
- [6] D. J. Nowak, E. J. Greenfield, R. E. Hoehn, and E. Lapoint, “Carbon Storage and Sequestration by Trees in Urban and Community Areas of the United States,” *Environmental Pollution*, vol. 178, pp. 229–236, 2013, doi: [10.1016/j.envpol.2013.03.019](https://doi.org/10.1016/j.envpol.2013.03.019).
- [7] European Environment Agency, “EMEP/EEA Air Pollutant Emission Inventory Guidebook 2023,” Copenhagen, EEA Technical Report, 2023. [Online]. Available: <https://www.eea.europa.eu/publications/emep-eea-guidebook-2023>
- [8] H. Hersbach *et al.*, “The ERA5 Global Reanalysis,” *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020, doi: [10.1002/qj.3803](https://doi.org/10.1002/qj.3803).
- [9] World Health Organization, “WHO Global Air Quality Guidelines,” Geneva, technical report, 2021. [Online]. Available: <https://www.who.int/publications/item/9789240034228>
- [10] Y. Ren, H. Liu, J. Wu, J. Li, and N. Zhang, “Nocturnal Boundary Layer Dynamics and Its Impact on Urban Air Quality in Tropical Megacities,” *Science of the Total Environment*, vol. 789, p. 147835, 2021, doi: [10.1016/j.scitotenv.2021.147835](https://doi.org/10.1016/j.scitotenv.2021.147835).

Student

LE MARREC Kelig

Study year in the department

4th year

Company

Hanoi University of Mining and Geology (HUMG)

Address

Dong Ngac, Bac Tu Liem,

Hanoi, Vietnam

Administrative manager

Dr. Nguyen Gia Trong, Lecturer-Researcher, Department of Higher Geodesy

nguyengiatrong@humg.edu.vn

Internship mentor

Eric Gascard

eric.gascard@grenoble-inp.fr

Referent teacher

Jean-François Méhaut

jean-francois.mehaut@univ-grenoble-alpes.fr

Title

Artificial Intelligence and GIS Applied to Urban Mobility Simulation and CO₂ Emissions Modelling

Summary

This report presents the work carried out during a 16-week internship (April 6 to July 27, 2026) at the Hanoi University of Mining and Geology (HUMG), Vietnam, under the supervision of Dr. Nguyen Gia Trong. The objective was to develop from scratch a simulation system for traffic-related urban CO₂ emissions, combining Artificial Intelligence and Geographic Information Systems (GIS).

The main deliverable is HUCODT (*Hanoi Urban CO₂ Digital Twin*), a fully open-source, Python-based urban air quality digital twin integrating four modules: YOLOv8-based vehicle detection, linear regression CO₂ prediction, an ERA5-forced atmospheric dispersion engine coupled with Oke's (1988) urban street-canyon model, and an interactive GIS visualisation layer built with Folium. Applied across 501 road segments around the HUMG campus, the system demonstrates that simulated nocturnal CO₂ concentrations (19.16 µg/m³) exceed afternoon values (14.87 µg/m³) despite a 65% traffic reduction, driven by the collapse of the planetary boundary layer height (PBLH) to 170 m. **Keywords:**

GIS, CO₂ emissions, urban traffic, digital twin, YOLOv8, ERA5, street canyon, Hanoi.